THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL UNIT TEST SET 1

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A right circular cylinder has base radius 7 cm and height 5 cm. The curved surface area of the cylinder is: 1

(a) 770 cm² **(b)** 220 cm² **(c)** 528 cm² **(d)** 440 cm²

2. The class interval is 10–20. The class mark is: 1

(a) 16 **(b)** 15 **(c)** 10 **(d)** 20

3. A die is thrown once. The probability of getting an even number is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{3}{7}$

4. A right circular cylinder has base radius 7 cm and height 10 cm. The volume of the cylinder is: 1

(a) 748 cm³ **(b)** 440 cm³ **(c)** 1540 cm³ **(d)** 3080 cm³

5. For the data with classes 10–20, 20–30, 30–40, 40–50 with frequencies 5, 8, 20, 17 respectively, The modal class is: 1

(a) 40–50 **(b)** 30–40 **(c)** 20–30 **(d)** 10–20

6. A die is thrown once. The probability of getting a multiple of 3 is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{3}$

7. A right circular cone has base radius 7 cm and height 24 cm. The slant height of the cone is: 1

(a) 25 cm **(b)** 31 cm **(c)** 23 cm **(d)** 15.5 cm

8. For the data with classes 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 7, 17, 20, 3, 3 respectively, The median class is: 1

(a) 30–40 **(b)** 40–50 **(c)** 50–60 **(d)** 20–30

9. A coin is tossed two times. The probability of getting two heads is: 1

(a) 0 **(b)** $\frac{1}{2}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{1}{4}$

10. A right circular cone has base radius 7 cm and height 24 cm. The curved surface area of the cone is: 1

(a) 528 cm² **(b)** 550 cm² **(c)** 704 cm² **(d)** 154 cm²

11. For the data with classes 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 3, 9, 4, 18, 16 respectively, The total number of observations $\sum f_i$ is: 1

(a) 49 **(b)** 47 **(c)** 51 **(d)** 50

12. A bag contains 3 red, 5 blue balls. One ball is drawn at random. The probability that it is red is: 1

(a) $\frac{5}{8}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{3}{8}$ **(d)** $\frac{1}{2}$

13. A solid sphere has radius 7 cm. The surface area of the sphere is: 1

(a) 154 cm² **(b)** 308 cm² **(c)** 462 cm² **(d)** 616 cm²

14. For the data with classes 0–10, 10–20, 20–30, 30–40 with frequencies 3, 9, 5, 8 respectively, The cumulative frequency of the class 10–20 is: 1

(a) 9 **(b)** 25 **(c)** 12 **(d)** 3

15. 10 cards numbered 1 to 10 are placed in a box, mixed well and one is drawn at random. The probability that the number drawn is a multiple of 3 is: 1

(a) $\frac{1}{5}$ **(b)** $\frac{2}{5}$ **(c)** $\frac{7}{10}$ **(d)** $\frac{3}{10}$

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A right circular cylinder has base radius 7 cm and height 10 cm. Find the total surface area of the cylinder. 2

17. The marks obtained by students in a test are given below. 2

Marks	10–20	20–30	30–40	40–50
Number of students	5	8	20	17

Find the mean.

18. A die is thrown once. Find the probability of getting a number greater than 6. 2

19. A solid hemisphere has radius 10.5 cm. Find the volume of the hemisphere. 2

20. The daily electricity consumption (in units) of households in a colony is given below. 2

Units consumed	40–50	50–60	60–70	70–80
Number of households	10	20	5	15

Find the mode.

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A right circular cone has base radius 7 cm and height 24 cm. Find the volume of the cone. 3

22. The marks obtained by students in a test are given below. 3

Marks	0–10	10–20	20–30	30–40	40–50
Number of students	12	20	8	4	6

Find the mean.

23. In a lottery there are 10 prizes and 25 blanks. A ticket is drawn at random. Find the probability of getting a prize. 3

24. A right circular cone has base radius 7 cm and height 24 cm. Find the total surface area of the cone. 3

OR

The daily wages (in ₹) of workers in a factory are given below. 3

Daily wage (₹)	20–30	30–40	40–50	50–60	60–70
Number of workers	5	16	5	13	11

Find the median.

25. A lot of 144 ball pens contains 20 defective pens. One pen is drawn at random from the lot. Find the probability that the pen is good (not defective). 3

OR

The total surface area of a cube is 54 cm^2. Find the volume of the cube. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. A solid metal cuboid 24 cm by 12 cm by 6 cm is melted and recast into cubes of edge 3 cm. Find the number of cubes obtained. 5

OR

The marks obtained by students in a test are given below. 5

Marks	50–60	60–70	70–80	80–90	90–100	100–110
Number of students	14	2	20	2	9	3

Find the mean.

27. A bag contains 3 red, 4 blue, 3 green balls. One ball is drawn at random. Find the probability that the ball is: 5

(i) red,

(ii) not blue,

(iii) yellow (in this order).

OR

A solid metallic sphere of radius 6 cm is melted and recast into small solid spheres of radius 2 cm. Find the number of small spheres obtained. 5

END OF QUESTION PAPER Total: 50 Marks

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